

Nathaniel Oliver

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EDUCATION

University of California, San Diego

M.S. Data Science, GPA: 4.0/4.0

La Jolla, CA

Expected June 2027

University of California, San Diego

B.S. Mathematics & Economics, Minor in Data Science

La Jolla, CA

June 2025

- *Relevant Coursework:* Machine Learning & Probabilistic Modeling, Representation Learning, Statistical Inference, Probability, Linear Algebra, Econometrics

EXPERIENCE

Data Science Intern

Jan. 2026 – Present

IDX Exchange

- Built an end-to-end **ML pipeline** predicting California single-family home prices from 100K+ **CRMLS** records – geospatial filtering, **MCAR/MAR/MNAR missingness analysis**, and imputation – with a **stacked ensemble (XGBoost + LightGBM)** reaching **7.74% MdAPE**, on par with Redfin ([live app](#))
- Designing a production **continuous-training MLOps pipeline** on AWS (**EventBridge** → **Lambda** → **S3** → **Glue** → **SageMaker Feature Store** → **Pipelines**): weekly CRMLS delta ingestion, **leakage-safe point-in-time features**, and monthly **XGBoost** retrains with time-decayed weighting and a champion-challenger gate
- Extending it into a **consumer-facing recommender (React/TS + FastAPI, Postgres)**: a content-based ranker (preference match + value score + freshness) plus an **LLM chatbot (Claude API tool-calling)** that captures user preferences, scored weekly via **Batch Transform**

Economic Research Assistant

Apr. 2024 – June 2025

UCSD Department of Economics

La Jolla, CA

- Built reproducible **ETL pipelines** in Python, R, and Stata to clean, aggregate, and explore large multi-source datasets (UNHCR, USCIS), documented for reuse by collaborating researchers
- Applied **time-series structural-break** (Bai-Perron, CUSUM/MOSUM) and **difference-in-differences** analysis to estimate effects across complex observational datasets, communicating findings to non-technical stakeholders

PROJECTS

Graph Neural Nets vs. Gradient Boosting ([repo](#)) | *PyTorch, PyTorch Geometric, RelBench* 2026 – Present

- Built a fair, **slice-aware** benchmark pitting a **heterogeneous two-tower GraphSAGE GNN (RelBench, NeurIPS 2024)** against a tuned **XGBoost LambdaRank** ranker for next-item recommendation on 218K Diginetica sessions – shared candidate pool, **budget-matched Optuna** – to isolate where graph structure beats flat features
- Ran a **relational-time leakage audit** that overturned an apparent GNN win: a seconds-resolution event clock was leaking each held-out item into the model's **time-aware receptive field**; fixed with **microsecond timestamps and last-prefix seeding** (verified by a subgraph-reachability probe), yielding the honest result that **XGBoost wins every slice (Novel Recall@10 0.174 vs. 0.103)**, matching the CIKM Cup 2016 winners

Agentic Resume Tailoring System (ARTie) ([repo](#)) | *Python, LangGraph, FastAPI* | [live demo](#) 2026 – Present

- Built a **multi-agent LangGraph pipeline** orchestrating **OpenAI and Anthropic LLMs** that ingests resumes, transcripts, LinkedIn profiles, and GitHub repos into a **knowledge graph**, with **multi-strategy skill matching** (lookup, fuzzy, graph traversal, embedding similarity) and a **reflection loop that prevents LLM fabrication**
- Drives resume tailoring through a **fast-path chat router** over tool-calling agents, backed by an **evaluation harness** that regression-tests routing and tailoring quality across scenarios
- Shipped end-to-end to **production**: React + TypeScript frontend and Python **FastAPI** backend with auth and Postgres, deployed on **Fly.io**, alongside a scripting **CLI**

TECHNICAL SKILLS

Languages & Libraries: Python, SQL, pandas, scikit-learn, PyTorch

LLM & AI: Retrieval-Augmented Generation (RAG), vector embeddings, Claude Code

MLOps & Cloud: AWS, PostgreSQL, Supabase, Git